

Python第9次(第10周)作业 (参考答案)

考试形式: 开卷

考试时间: 2024-4-29

院系: 东吴学院 年级: 2023 专业: 非计算机专业

学号: _____ 姓名: _____ 分数: _____

一、选择题 (每小题1.0分, 共20.0分)

- | | | | | |
|-------|-------|-------|-------|-------|
| 01. D | 02. B | 03. B | 04. D | 05. B |
| 06. C | 07. C | 08. C | 09. A | 10. B |
| 11. D | 12. D | 13. D | 14. C | 15. B |
| 16. B | 17. B | 18. B | 19. A | 20. D |

二、填空题 (每空2.0分, 共20.0分)

01. -2
02. 6
03. 1
04. None
05. 4 0
06. ['H', 'ppy birthd', 'y to you!']
07. 20
08. 2 3 1
09. a, b=b, a+b[或]a, b=b, b+a
10. line[j]+line[j+1][或]line[j]+line[1+j][或]line[j+1]+line[j][或]line[1+j]+line[j]

三、编程题 (每小题6.0分, 共60.0分)

01. (6.0分) 答:

```
def demo(m, n):  
    if m>n:  
        m, n = n, m  
    p = m*n  
    while m!=0:  
        r = n%m  
        n = m  
        m = r  
    return (n, p//n)
```

a,b=eval(input())

print(demo(a,b))

02. (6.0分) 答:

```
def total_nums(n):  
    s = 0
```

```

while n:
    s = s + n%10
    n = n//10
return s
def lucky_pair(m, n):
    if abs(m-n)==3 and total_nums(m)%6==0 and total_nums(n)%6==0:
        return True
    else:
        return False
m,n=eval(input())
if lucky_pair(m,n):
    print('Yes')
else:
    print('No')
03. (6.0分) 答:
def fai(m):
    s = 0
    for i in range(1, m + 1):
        if m%i==0:
            s = s + i
    return s
m = eval(input())
print(fai(m))
04. (6.0分) 答:
def trans(a):
    b = [[] for i in range(len(a[0]))]
    for i in range(len(b)):
        for j in range(len(a)):
            b[i].append(a[j][i])
    return b
x = eval(input())
y = trans(x)
for i in range(len(y)):
    for item in y[i]:
        print('%5d'%item, end=' ')
    print()
05. (6.0分) 答:
def prime(n):
    for i in range(2,n):
        if n%i==0:
            return False
    return True
n = int(input())

```

```

for i in range(2, n//2+1):
    if prime(i) and prime(n-i):
        print(' {}={}+{}'.format(n, i, n-i))

```

06. (6.0分) 答:

```

a,b,c = map(int, input().split())
f = lambda x,y: x if x>y else y
print(f(f(a,b),c))

```

07. (6.0分) 答:

```

def jc(n):
    result=1
    for i in range(1,n+1):
        result*=i
    return result
m,n=map(eval,input().split())
if m>=n>0:
    c=jc(m)/(jc(n)*jc(m-n))
    print('%d'%c)

```

else:

```

    print('Error')

```

08. (6.0分) 答:

```

def huiw(n):
    s = str(n)
    if s==s[::-1]:
        return True
    else:
        return False

```

```

m, n = eval(input())

```

if m>n:

```

    m, n = n, m

```

c = 0

```

for i in range(m, n+1):

```

```

    if huiw(i):

```

```

        print('%5d'%i, end='')

```

```

        c = c + 1

```

if c==0:

```

    print('No such number')

```

09. (6.0分) 答:

```

import math

```

```

def jf(a,b):

```

```

    if a>b:

```

```

        a,b = b,a

```

```

    h = (b-a)/1000

```

```

    s = 0

```

```

    for i in range(1,1000):
        s = s + math.sin(a+i*h)
    s = s + (math.sin(a)+math.sin(b))/2
    return s*h
a, b = eval(input())
print('%.3f'%jf(a,b))
10. (6.0分) 答:
def isNarcissistic(n):
    a=n//100
    b=n//10%10
    c=n%10
    if n==a*a*a+b*b*b+c*c*c:
        return True
    else:
        return False
m,n=eval(input())
if m>n:
    m,n=n,m
c = 0
for i in range(m,n+1):
    if isNarcissistic(i):
        print('%5d'%i, end='')
        c = c + 1
if c==0:
    print(None)

```